Fifa World Cup 2026

The Fifa World Cup is an international football tournament that started in 1930. From relatively small beginnings, it has grown to be a multi-billion dollar industry. The Fifa clearing house managing international player transfers saw nearly USD$1 billion since late 2022 (FIFA, 2026a). From 6.9 million fans at live matches to 29 billion video views on official social media (FIFA, 2026b), revenue from Fifa World Cup 2026 events including ticketing, marketing, broadcasting and sponsorship rights exceeded USD$9 billion (The Data Expert, 2026). These are just the Fifa numbers but it's the gambling industry where the big numbers lie. Gaming Compliance International reports USD$593 billion dollars as a conservative estimate of the wagering value of the 2026 competition (Sobrado, 2026). Statisticians record and analyse every conceivable statistic about each player, club and match. Predictive modelling is used to forecast outcomes and set odds for competitive gambling on the competition results.

This project attempts to predict goal difference in all 104 matches. It uses a multiple linear regression model with 8 explanatory variables. The variables chosen are based on pre-match data and based on coach salaries, official Fifa ranking, round number, statistical experience and age of each 26 player, and speculated expected goals per match.

Model 2.1 Multiple linear regression model to predict Goal difference

Variables

11 initial variables chosen to be reduced to 8 most effective. Each variable is the difference between teams except for round number.

1) Coach salary (this as indicative of an investment or a 'drive' within the team)
2) Ranking (as official recognised team skill level)
3) XG (expected goals by 'statsdontlie' statisticians)
4) Median age (may indicate experience level)

Caps – experience at international level. Grouped and averaged by playing position because teams differ in their makeup but there are only so many positions on the pitch at the same time

5) Goalkeeper cap (level of experience at professional level. Not averaged because all teams have 3)
6) Forward players mean cap
7) Defence players mean cap
8) Midfield players mean cap

Goals – by players at international level

9) Forward players mean goals
10) Squad goal rate (total goals divided by caps)
11) Round number (categorised from 1-8)

Data sources

https://worldcupstats.football/teams/#sort=delta_points%2Cdesc Team ranking at 1/04/2026

https://www.salaryleaks.com/blogs/world-cup-2026-coach-salary-rankings Coach estimated annual salary at 12/06/2026

https://fdp.fifa.org/assetspublic/ce281/pdf/SquadLists-English.pdf https://en.wikipedia.org/wiki/2026_FIFA_World_Cup_squads# Final squad stats 2/06/2026 caps, goals, players, DOB. PDF imported and cleaned in excel

https://www.thestatsdontlie.com/football/world-cup-2026/ Expected goals per match extracted from frame info embedded google sheet

https://theanalyst.com/articles/france-vs-england-stats-world-cup-2026-bronze- Expected goal for the third place final (not listed on statsdontlie)

Methods

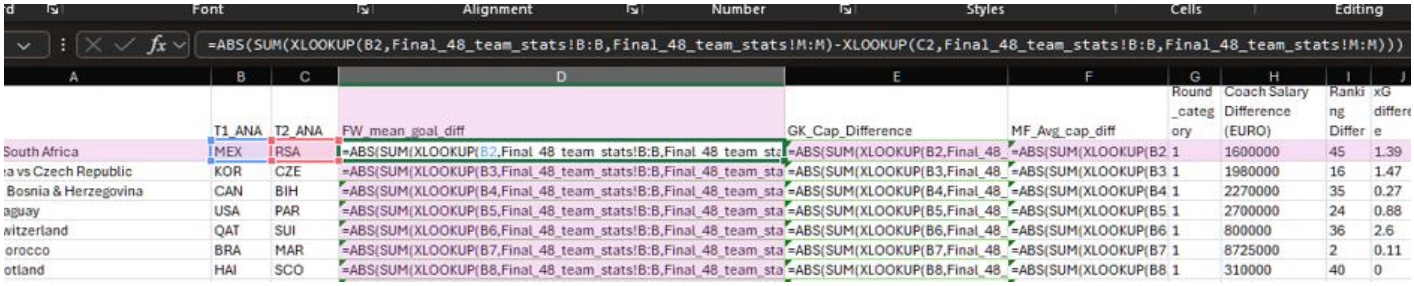
Once the variables were decided on and the data located, the data was loaded into excel for cleaning and collating. Raw player data was grouped and categorised in python(figures1,2) before exporting back to excel to finalise the linear regression datasheet (figure 3). Initial testing used 11 variables. After evaluating model power and correlation (figure 4), this number was reduced to 8 by removing the variables with least relationship to actual goal difference. Surprisingly squad goal rate, midfield and defence player caps had less impact on goal difference than coach salary so these were eliminated in the final dataset.

```
Finalised Fifa squad: raw player data
<class 'pandas.DataFrame'>
RangeIndex: 1248 entries, 0 to 1247
Data columns (total 11 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Country_ANA    1248 non-null   str
 1   Country        1248 non-null   str
 2   #              1248 non-null   int64
 3   POS            1248 non-null   str
 4   PLAYER NAME    1248 non-null   str
 5   NAME ON SHIRT  1248 non-null   str
 6   DOB            1248 non-null   int64
 7   CLUB           1247 non-null   str
 8   HEIGHT (CM)    1248 non-null   int64
 9   CAPS           1248 non-null   int64
 10  GOALS          1248 non-null   int64
dtypes: int64(5), str(6)
memory usage: 107.4 KB
None
```

Figure 1 Raw player dataset

```
Final squad player data grouped into countries,
 caps and goals calculated by playing positions,
 median team age at 11/06/2026 calculated from DOB
<class 'pandas.DataFrame'>
RangeIndex: 48 entries, 0 to 47
Data columns (total 14 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Country_ANA     48 non-null     str
 1   CAPS_total      48 non-null     int64
 2   GOALS_total     48 non-null     int64
 3   CAPS_average    48 non-null     float64
 4   GOALS_average   48 non-null     float64
 5   Age_median      48 non-null     float64
 6   CAPS_DF_mean    48 non-null     float64
 7   CAPS_FW_mean    48 non-null     float64
 8   CAPS_GK_mean    48 non-null     float64
 9   CAPS_MF_mean    48 non-null     float64
 10  GOALS_DF_mean   48 non-null     float64
 11  GOALS_FW_mean   48 non-null     float64
 12  GOALS_GK_mean   48 non-null     float64
 13  GOALS_MF_mean   48 non-null     float64
dtypes: float64(11), int64(2), str(1)
```

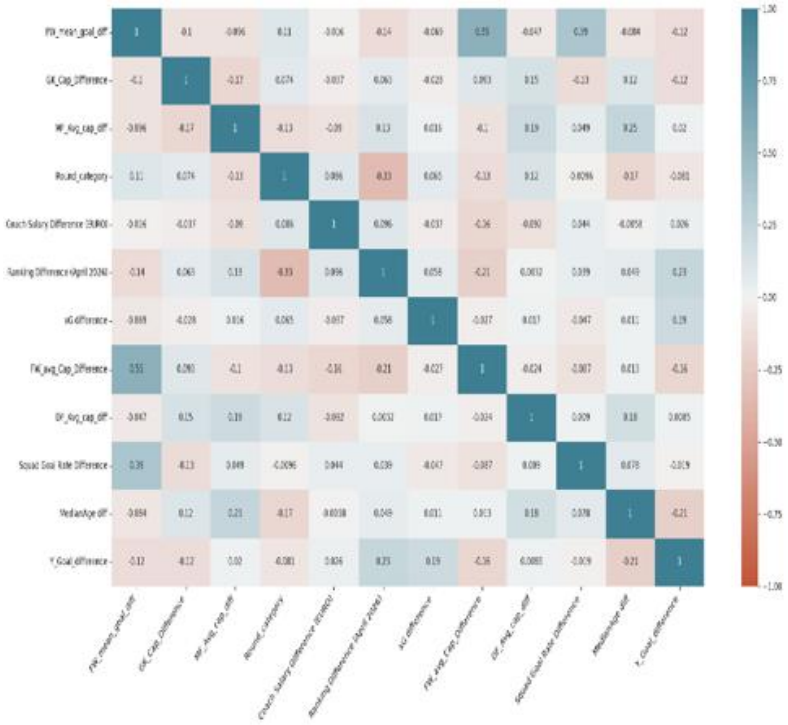
Figure 2 Squad dataset from player data



=ABS(SUM(XLOOKUP(B2,Final_48_team_stats!B:B,Final_48_team_stats!M:M)-XLOOKUP(C2,Final_48_team_stats!B:B,Final_48_team_stats!M:M)))

	T1_ANA	T2_ANA	FW_mean_goal_diff	GK_Cap_Difference	MF_Avg_cap_diff	Round_category	Coach Salary Difference (EURO)	Ranking Difference	xG difference
South Africa	MEX	RSA	=ABS(SUM(XLOOKUP(B2,Final_48_team_stats!B:B,Final_48_team_sta	=ABS(SUM(XLOOKUP(B2,Final_48_	=ABS(SUM(XLOOKUP(B2	1	1600000	45	1.39
a vs Czech Republic	KOR	CZE	=ABS(SUM(XLOOKUP(B3,Final_48_team_stats!B:B,Final_48_team_sta	=ABS(SUM(XLOOKUP(B3,Final_48_	=ABS(SUM(XLOOKUP(B3	1	1980000	16	1.47
Bosnia & Herzegovina	CAN	BIH	=ABS(SUM(XLOOKUP(B4,Final_48_team_stats!B:B,Final_48_team_sta	=ABS(SUM(XLOOKUP(B4,Final_48_	=ABS(SUM(XLOOKUP(B4	1	2270000	35	0.27
aguay	USA	PAR	=ABS(SUM(XLOOKUP(B5,Final_48_team_stats!B:B,Final_48_team_sta	=ABS(SUM(XLOOKUP(B5,Final_48_	=ABS(SUM(XLOOKUP(B5	1	2700000	24	0.88
witzerland	QAT	SUI	=ABS(SUM(XLOOKUP(B6,Final_48_team_stats!B:B,Final_48_team_sta	=ABS(SUM(XLOOKUP(B6,Final_48_	=ABS(SUM(XLOOKUP(B6	1	800000	36	2.6
orocco	BRA	MAR	=ABS(SUM(XLOOKUP(B7,Final_48_team_stats!B:B,Final_48_team_sta	=ABS(SUM(XLOOKUP(B7,Final_48_	=ABS(SUM(XLOOKUP(B7	1	8725000	2	0.11
otland	HAI	SCO	=ABS(SUM(XLOOKUP(B8,Final_48_team_stats!B:B,Final_48_team_sta	=ABS(SUM(XLOOKUP(B8,Final_48_	=ABS(SUM(XLOOKUP(B8	1	310000	40	0

^ *Figure 3 Data wrangling in excel*



< *Figure 4 Correlation heatmap-11 variables*

Results

```
Multiple Linear Regression performance metrics:
   data scaled : {StandardScaler()}
   (train/test split at 0.20 test)
MAE:  1.192
MSE: 2.891
RMSE: 1.700
RMSE (Normalised): 0.283
R^2: -0.025

 Model perferformance at 0.20 test split,
  using (10)  Kfold validation
Average   R^2   : -0.038
  Average NRMSE : 0.216
```

Figure 5 Skicit-learn model performance metrics

Multiple scalers were tried in Python to find best fit for modelling but there appeared to be no strong linear correlations in the dataset. Scikit-Learn linear regression with StandardScaler and K-fold validation performed very poorly with negative results (figure 5) but at least better than Baseline modelling (figure 6). The best combination was Statsmodel package with PowerTransform scaler. With an adjusted R^2 = .1, this model can explain 10% of results.

```
##### BASELINE MODEL #####
(based on mean data values)
   Actual  Predicted
0       1   1.585106
1       0   1.585106
2       2   1.585106
3       1   1.585106
4       1   1.585106
MAE: 0.617
MSE: 0.491
RMSE: 0.701
RMSE (Normalised): 0.117
R^2:  -0.198
```

^*Figure 6 Baseline performance metrics*

```
                            OLS Regression Results
==============================================================================
Dep. Variable:      Y_Goal_difference   R-squared:                       0.170
Model:                            OLS   Adj. R-squared:                  0.100
Method:                 Least Squares   F-statistic:                     2.433
Date:                Wed, 14 Oct 2026   Prob (F-statistic):             0.0194
Time:                        16:24:37   Log-Likelihood:                -170.21
No. Observations:                 104   AIC:                             358.4
Df Residuals:                      95   BIC:                             382.2
Df Model:                           8
Covariance Type:            nonrobust
================================================================================================
                                   coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------------------
const                            1.5577      0.128     12.213      0.000       1.304       1.811
FW_mean_goal_diff               -0.0605      0.146     -0.413      0.681      -0.351       0.230
GK_Cap_Difference               -0.1698      0.135     -1.257      0.212      -0.438       0.098
Round_category                  -0.0641      0.145     -0.441      0.660      -0.353       0.225
Coach Salary Difference (EURO)  -0.0960      0.131     -0.734      0.465      -0.356       0.164
Ranking Difference (April 2026)  0.2914      0.143      2.037      0.044       0.007       0.575
xG difference                    0.2359      0.129      1.832      0.070      -0.020       0.492
FW_avg_Cap_Difference           -0.1600      0.150     -1.064      0.290      -0.458       0.139
MedianAge diff                  -0.2657      0.133     -2.005      0.048      -0.529      -0.003
==============================================================================
Omnibus:                       13.141   Durbin-Watson:                   2.070
Prob(Omnibus):                  0.001   Jarque-Bera (JB):               14.750
Skew:                           0.745   Prob(JB):                     0.000627
Kurtosis:                       4.088   Cond. No.                         2.01
==============================================================================
```

< Figure 4 Multiple linear regression summary after power transformation

Conclusion

These variables are ultimately not very effective at predicting goal difference. I believe this could be improved upon using different variables, perhaps player possession or shot success rates. Top player salary might also show interesting results considering the effect of coach salary. This process has illustrated the important of accessing expert knowledge and doing rigorous research earlier in the project to choose appropriate explanatory variables.

References

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